



When Gods Bleed

Total War: Warhammer III
Modlist & Guide

Version 8.1.1 — July 2026

RPFM Mod Manager & Typst Documentation

Table of Contents

Contents

Table of Contents	2
Installation Guide	5
RPFM Setup (Recommended)	5
RPFM Setup for When Gods Bleed	5
Overview	5
Installation	5
First-Time Configuration	5
Step 1: Set the Game Directory	5
Step 2: Configure a Mod Folder	5
Activating Mods from This Modlist	5
Downloading Mods from URLs	5
Recommended Workflow for This Modlist	6
Troubleshooting	6
Wave 0 Installation	6
Wave 0 — Modlist: Vanilla + UI/QoL	6
UI & Quality of Life	6
Notes	7
Wave 1 Installation	7
Wave 1 — Modlist: Content	7
Required Framework	7
New Factions	8
New Legendary Lords & Characters	8
Notes	9
Wave 2 Installation	9
Wave 2 — Modlist: Mechanics	9
Total Overhaul	9
Campaign Systems	9
Army & Battle	10
Notes	10
Wave 3 Installation	10
Wave 3 — Modlist: Full Chaos	10
UI / QoL (Wave 0 Carryover)	11
Graphics (Waves 0–2 Carryover)	11
Notes	12
Mod Activation Order	13
Conflicts & Exclusions	13
Waves & Campaign Setups	14
Wave 0: The Old World	14
Roleplaying Background	14
How to Play — Wave 0 (Beginner)	14
Campaign Setup	14

Core Mechanics to Master	14
Economy	14
Military	14
Diplomacy	14
Turn Flow	14
Recommended Progression	15
When You Are Ready for the Next Wave	15
Wave 0 — Modlist: Vanilla + UI/QoL	15
UI & Quality of Life	15
Notes	16
Wave 0 — Graphics Mods	16
Notes	17
Wave 0 — Content Mods	17
Notes	17
Wave 0 — Mechanics Mods	17
Wave 1: The Gathering Storm	18
Roleplaying Background	18
Wave 1 Strategy Guide — The Gathering Storm	18
Campaign Philosophy	18
Faction Strategy Adjustments	18
Empire	18
Other Factions	18
Battle Tactics	18
Economy in Wave 1	18
Readiness for Wave 2	19
Wave 1 — Modlist: Content	19
Required Framework	19
New Factions	19
New Legendary Lords & Characters	20
Notes	20
Wave 1 — Graphics Mods	20
Notes	21
Wave 1 — Content Mods: World & Characters	21
Notes	21
Wave 1 — Mechanics Mods	22
Wave 2: The End Times	23
Roleplaying Background	23
Wave 2 Guide — The End Times	23
Campaign Philosophy	23
Core Mechanics Changes	23
SFO Combat Pacing	23
Victory Routes	23
Army Composition Limits	23
Smarter AI	23
Difficulty Curve	24
Recommended Progression	24
Prerequisites Before Wave 3	24
Important	24

Wave 2 — Modlist: Mechanics	24
Total Overhaul	24
Campaign Systems	25
Army & Battle	25
Notes	25
Wave 2 — Graphics Mods	26
Notes	26
Wave 2 — Content Mods: SFO Integration	26
Notes	27
Wave 2 — Mechanics Mods (continued)	27
Notes	27
Wave 3: When Gods Bleed	28
Roleplaying Background	28
Wave 3 — When Gods Bleed	28
Campaign Philosophy	28
Combined Mod Impact	28
Difficulty Expectations	28
Recommended Approach	28
Final Note	29
Wave 3 — Modlist: Full Chaos	29
UI / QoL (Wave 0 Carryover)	29
Graphics (Waves 0–2 Carryover)	30
Notes	31
Wave 3 — Graphics Mods	31
Notes	31
Wave 3 — Content Mods	31
Wave 3 — Mechanics Mods	32
Final Note	32

Installation Guide

RPFM Setup (Recommended)

RPFM Setup for When Gods Bleed

Overview

link(<https://frodo45127.github.io/rpfm/>)[RPFM] is the official mod manager for Total War: Warhammer III. This guide walks you through the full setup process required for the When Gods Bleed modlist.

Installation

1. Download the latest release from the link(<https://github.com/frodo45127/rpfm/releases>)[RPFM GitHub Releases].
2. Extract the archive to a permanent location (e.g., C:\RPFM).
3. Launch RPFM.exe — it will attempt to auto-detect your TWWH3 installation.

First-Time Configuration

Step 1: Set the Game Directory

1. Open RPFM and click the **Settings** gear icon.
2. Locate the **Warhammer 3 Directory** field.
3. Set it to your TWWH3 root folder. Example paths:
 - **Steam:** C:\Program Files\Steam\steamapps\common\Total War WARHAMMER 3
 - **Epic Games Store:** C:\Program Files\Epic Games\TWW3
4. Click **Save** or **Apply**.

Step 2: Configure a Mod Folder

1. In Settings, go to the **Mods** tab.
2. Set a dedicated mod folder path for this modlist:
 - Recommended: C:\TWWH3\Mods\WhenGodsBleed
3. This keeps Wave-specific mods organized and prevents conflicts with other modlists.
4. Click **Save**.

Activating Mods from This Modlist

1. In RPFM, navigate to **Mods** in the left sidebar.
2. Click **Add Mod** and browse to the mod file or folder.
3. Toggle the mod to **Active** (green checkmark icon).
4. Verify dependency chains: any mod listed as a dependency must also be active.
5. Click **Launch Game** to start TWWH3 with your selected mods.

Downloading Mods from URLs

1. Click **Add Mod** > **Download from URL**.
2. Paste the mod URL (e.g., a Total Workshop link).
3. RPFM downloads and adds the mod to your library.
4. Activate it and check dependencies as above.

Recommended Workflow for This Modlist

1. Start by installing and activating all **Wave 0** mods.
2. Launch the game and complete the campaign tutorial / learn basic mechanics.
3. When ready for more challenge, deactivate Wave 0 mods (optional) and activate Wave 1 mods.
4. Continue wave progression at your own pace.
5. Never activate mods from multiple waves simultaneously unless specified otherwise.

Troubleshooting

| Symptom | Likely Cause | Fix | |———|———-|——| | RPFM cannot find the game | TWWH3 directory not set | Manually set the path in Settings | | Mod crashes on launch | Version mismatch or missing dependency | Check mod version and activate all dependencies | | Conflicting mods cause issues | Two mods modify the same content | Refer to conflicts.conf; deactivate one | | Game won't start after modding | Mod order or activation issue | Deactivate all mods, reactivate in wave order |

Wave 0 Installation

Wave 0 — Modlist: Vanilla + UI/QoL

This wave contains only vanilla-compatible and QoL UI mods. No gameplay-altering mechanics are included.

UI & Quality of Life

Better Camera Mod

Version: v8.1 (Jul 2026)

Impact: Unlocks closer and further camera zoom heights in battles and on the campaign map, and lets you zoom without units popping out of view. Makes battles feel more cinematic and easier to watch. Purely camera QoL with no gameplay change.

Cpecific's Skill Queue: Jazzed Edition

Version: v8.1 (Jul 2026)

Dependencies: Memreader (win64)

Impact: Lets you build skill templates and auto-assign skill points to every new lord and hero as they level up — never hand-click a skill tree again. The community-maintained fork of the original Skill Queue, with saved-game and faction templates plus auto-granted-skill detection. Saves hours; gives no gameplay advantage.

Memreader (win64)

Version: v8.1

Impact: Required memory-access library for Skill Queue and similar QoL mods. Windows x64 only. Install before any mod that lists it as a dependency.

The Tooltip Project

Version: v8.1 (Jul 2026)

Impact: Rewrites unit and spell tooltips so they tell you what actually matters: collision attacks, melee regeneration, hidden stat mechanics, and whether a buff spell is single- or multi-target. Information-only UI clarity — the author states it will never touch gameplay.

Building Progression Icons III

Version: v8.1 (Dec 2025)

Impact: Replaces flat building icons with artwork that visibly changes as each building is upgraded, so settlement state is readable at a glance. Purely cosmetic.

Speedy Turn Camera

Version: v8.1 (May 2026)

Impact: Speeds up the camera pan between factions during the AI's turn, so long end turns feel much faster. Changes a single database value — it does not speed up AI processing, just the camera travel.

High Resolution UI Improvements, YES IT STILL WORKS

Version: v8.1 (Dec 2025)

Impact: Resizes the recruitment, diplomacy, lord/hero, objectives, army-view and custom-battle panels so they display correctly on 1440p+ monitors. Note: some users report the recruitment panel can lag or freeze on 3440x1440 ultrawide.

Skip Intro

Version: v8.1 (May 2026)

Impact: Skips the startup intro videos and the epileptic-seizure warning at launch. Small convenience; be aware that running any mod adds a "Modified Game" confirmation popup at launch.

Notes

Wave 0 is designed for new players. The mods here only improve the interface and quality of life — nothing changes core game balance or mechanics. Skill Queue requires Memreader; install both or the preset features will misbehave.

Wave 1 Installation

Wave 1 — Modlist: Content

This wave expands the game with new factions, units, legendary characters, and campaign features. Everything in this wave builds on the MIXER startpos framework, so install MIXER first and keep every dependency chain active.

Required Framework

MIXER - Mixu's Unlocker

Version: v8.1 (Jul 2026)

Dependencies: Mod Configuration Tool (recommended)

Impact: The startpos unlocker that makes all factions playable (458 lords in Immortal Empires) and the modder framework nearly every content mod in this wave is built on. Activate this first — content mods will not function without it. Note: single-player only for now; Aug 2026 CA update broke multiplayer.

New Factions

Cataph's Southern Realms (TEB) [It Works in 8.1]

Version: v8.1 (Dec 2025)

Dependencies: MIXER - Mixu's Unlocker; free WH3 Jade/Grey/Amethyst Wizard packs

Impact: Turns Tilea, Estalia, the Border Princes and New World Colonies into a full culture: 8 playable factions, 10 legendary lords and 1 legendary hero, 30+ new units, custom tech and building trees, landmarks, and the Plutocracy and mercenary-camp mechanics. Not for the Realms of Chaos campaign.

OvN Lost Factions: Fimir

Version: v8.1 (Jul 2026)

Dependencies: MIXER - Mixu's Unlocker (required); not multiplayer-campaign compatible

Impact: Adds the Fimir as a complete new playable race: full custom roster, 2 playable factions (Tendrils of Doom, Rancor Hold), and custom mechanics including Godhood Ascension, Gifts of Chaos and the Death Quest.

Cataph's Kraka Drak: the Norse Dwarfs [It Works in 8.1]

Version: v8.1 (Dec 2025)

Dependencies: MIXER - Mixu's Unlocker (required)

Impact: Adds the Norse Dwarf sub-faction under Great King Thorgard Cromson: 11+ new units, 8 Regiments of Renown, a custom legendary lord type, and new tech tree, buildings and landmarks. For Immortal Empires / The Old World campaigns.

New Legendary Lords & Characters

Mixu's Legendary Lords

Version: v8.1 (Jul 2026)

Dependencies: Mixu's Legendary Lords: Asset Pack

Impact: The community-standard legendary lord pack: 50+ lore-accurate legendary lords and heroes across existing races, with custom skill trees and new start positions. MIXER is recommended to play the added lords.

Mixu's Legendary Lords: Asset Pack

Version: v8.1 (Jul 2026)

Impact: Required asset pack for Mixu's Legendary Lords — without it you will get missing character models and other broken assets. Install alongside the main mod.

[DEER24] Walk with the Dragon (Cathay Units Patch)

Version: v8.1 (Aug 2026)

Dependencies: Mixu's faction unlocker (required); optional English-patch submod

Impact: Cathay mega-pack: 14 legendary lords, 11 legendary heroes, 2 new hero types, 60+ new units and a 50-turn event chain. Chinese-language text by default (English patch submod exists). Immortal Empires only.

Notes

Wave 1 mods add significant content. MIXER is the mandatory hub — activate it before anything else. Mixu's Legendary Lords requires its Asset Pack. All mods here are single-player focused; check conflicts.conf for startpos interactions (e.g. Immortal Empires Expanded + Landmarks of Eternity pairing is covered in the content section).

Wave 2 Installation

Wave 2 — Modlist: Mechanics

This wave introduces mechanics overhaul mods that change core gameplay systems. SFO: Grimhammer III is the foundation — every other mechanic in this wave is chosen to work with it. Start a new campaign when activating this wave.

Total Overhaul

SFO: Grimhammer III - Call of Bretonnia

Version: v8.1 (Jul 2026)

Dependencies: None required; fully standalone

Impact: The flagship lore-flavoured total overhaul: longer battles, faction-mechanic reworks, slower XP progression, optional unit caps, and everything rebalanced against the tabletop-inspired vision. Completely replaces vanilla balance — this is the single biggest change in the whole modlist.

Campaign Systems

Victory Conditions Overhaul

Version: v8.1 (Jul 2026)

Dependencies: VCO - SFO Compatibility Patch (required with SFO)

Impact: Replaces the short/long victory objectives with 3 thematic routes per faction (50–70 turns), each with unique rewards, quest battles and epilogues. Reshapes what "winning" means. Must be enabled at campaign start — objectives are set on turn 1.

Diplomacy Logic and Balance Katsuko

Version: v8.1 (Dec 2025)

Dependencies: None; stated compatible with all mods incl. SFO

Impact: Rebalances every core diplomatic factor: distance-aware war declarations, no instant trust between enemies, selling/gifting/purchasing settlements, and dynamic alliances. Diplomacy becomes logical and strategic instead of arbitrary.

Army & Battle

Tabletop Caps: Reborn

Version: v8.1 (Dec 2025)

Dependencies: Optional: Mod Configuration Tool for tuning

Impact: Tabletop-style Core/Special/Rare unit caps per army, applied to the player AND the AI. Kills doomstacks and forces balanced, lore-like army compositions. The primary army-cap system for this wave.

AI Recruitment & Army Compositions (8.1.1)

Version: v8.1.1 (Jul 2026)

Dependencies: None; incompatible with mods touching `cdir_military_generator`

Impact: Rewrites how the AI builds armies: more line infantry, capped ranged/artillery/monsters, and themed templates per legendary lord. AI fielding balanced armies instead of spam. Incompatible with Campaign AI Tweaks — do not run both.

Siege AI & Mechanics Overhaul

Version: v8.1 (May 2026)

Dependencies: None; compatible with most garrison mods

Impact: Reworks sieges: the AI attacks instead of attrition-starving, defensive buildings grant attrition/ammo/upkeep bonuses, no post-battle tower building, healthier gates and walls, and reworked settlement battle types.

Notes

Wave 2 mechanics mods are deep gameplay changes. Activate SFO first, then the rest. The VCO-SFO compatibility patch and SFO content patches for Wave 1 mods are listed in the mechanics and content sections — read those before launching. Never pair SFO with Radium Total War (see `conflicts.conf`). Start a fresh campaign.

Wave 3 Installation

Wave 3 — Modlist: Full Chaos

This wave activates ALL mods from Waves 0–3 simultaneously. This is the complete modlist. Activate in wave order: QoL (below), graphics (below + Wave 3 graphics section), then content and mechanics from the Wave 1/2 sections.

UI / QoL (Wave 0 Carryover)

Better Camera Mod

Version: v8.1 (Jul 2026)

Impact: Unlocked camera heights in battles and campaign; zoom without unit culling.

Cpecific's Skill Queue: Jazzed Edition

Version: v8.1 (Jul 2026)

Dependencies: Memreader (win64)

Impact: Skill presets that auto-assign points to new lords/heroes.

Memreader (win64)

Version: v8.1

Impact: Memory library required by Skill Queue.

The Tooltip Project

Version: v8.1 (Jul 2026)

Impact: Rewrites unit/spell tooltips with hidden stats and mechanics.

Building Progression Icons III

Version: v8.1 (Dec 2025)

Impact: Building icons change as buildings upgrade.

Speedy Turn Camera

Version: v8.1 (May 2026)

Impact: Faster end-turn camera pan between factions.

High Resolution UI Improvements

Version: v8.1 (Dec 2025)

Impact: Resizes UI panels for 1440p+ monitors.

Skip Intro

Version: v8.1 (May 2026)

Impact: Skips startup intros and the epilepsy warning.

Graphics (Waves 0–2 Carryover)

Colourless' Radious HD Texture Pack

Version: v8.1 (Jul 2026)

Impact: Comprehensive HD texture replacement across all base races.

Warhammer 1 Texture Update Project

Version: v8.1

Impact: Upgrades WH1 legacy race textures to WH3 standards.

Campaign Map Lighting

Version: v8.1

Impact: Warmer campaign map lighting with deeper shadows and day/night contrast.

ULTIMATE LIGHTING 2.0

Version: v2.0

Impact: Complete battle lighting overhaul with deeper shadows and better ambient occlusion.

Details Matter: True Grit Textures (The Empire)

Version: v8.1

Impact: High-detail Empire texture upgrade with realistic wear and battle damage.

Lore Soldiers: Units Reskin as Miniatures and Books

Version: v8.1

Impact: Tabletop-miniature-accurate reskins across multiple factions.

Grim Lighting

Version: v8.1

Impact: Dark, oppressive battle lighting with reduced saturation — the End Times look.

Chaos Visual Overhaul

Version: v8.1

Impact: Comprehensive visual overhaul for all Chaos factions.

Better Chaos Waste Lighting Effect

Version: v8.1

Dependencies: Chaos Visual Overhaul

Impact: Unique atmospheric lighting for Chaos Waste regions.

Notes

Wave 3 is the final wave. All mods from Waves 0–3 are active. Respect all dependency chains: Skill Queue needs Memreader, VCO needs its SFO patch, Wave 1 content needs MIXER and its SFO patches. Check conflicts.conf before activation — some mods may conflict even if not within the same wave.

Mod Activation Order

Activate mods in wave order (Wave 0 first, then Wave 1, Wave 2, Wave 3). Each wave builds on the previous wave's setup. Do not activate mods from multiple waves simultaneously unless explicitly noted.

Conflicts & Exclusions

Conflicting mods are listed in `conflicts.conf` and are excluded from this guide.

Waves & Campaign Setups

Wave 0: The Old World

Roleplaying Background

You are the newly appointed Governor of Averheim's outer provinces — a backwater region of the Empire long neglected by the Elector Counts. The old order still holds, but shadows gather on the northern horizon. Greenskins mass in the forests undetected, and something stirs beneath the mountains. You are given command of a small garrison and a meagre treasury. Your mission is simple: survive, learn, and prove yourself worthy of greater responsibilities.

This is your starting point. No allies, no legendary lords, no advantages — only your wits and whatever the Empire can provide.

How to Play — Wave 0 (Beginner)

Campaign Setup

When starting a new campaign in Wave 0, follow these guidelines:

1. **Choose the Empire** as your faction. It is the most forgiving faction for newcomers with balanced armies and a stable economy.
2. **Start on a Normal campaign map.** No extreme world settings.
3. **Do not use custom lords.** Use the pre-built Empire characters only.
4. **Set difficulty to Normal or Easy** until you are comfortable with mechanics.

Core Mechanics to Master

Economy

- Collect taxes from provinces each turn.
- Build infrastructure (roads, mines, farms) before military structures.
- Balance army maintenance costs with income — over-extending leads to bankruptcy.

Military

- Recruit basic Empire units: Swordsmen, Spearmen, Crossbowmen, and Cavalry.
- A balanced army should have: frontline infantry, ranged support, and a cavalry flanking unit.
- Do not engage enemies significantly stronger than your army.

Diplomacy

- Maintain peace with neighbouring factions early on.
- Trade resources to build income.
- Send diplomatic agents to strengthen relations.

Turn Flow

1. **Move agents** to explore, gather intelligence, or complete missions.
2. **Move armies** to secure border regions.
3. **Build** in each province — prioritize economy over military in peacetime.
4. **End turn** and review events.

Recommended Progression

- Turns 1-10: Expand cautiously to 2-3 provinces. Focus on economy.
- Turns 11-25: Build a moderate army. Deal with any local threats.
- Turns 26+: Explore neighbouring regions. Begin diplomacy with nearby factions.

When You Are Ready for the Next Wave

You should feel comfortable with:

- Managing economy and army upkeep
- Basic battle tactics
- Diplomatic relationships
- Understanding faction strengths and weaknesses

Once these feel natural, move on to Wave 1 for content additions and more complexity.

Wave 0 — Modlist: Vanilla + UI/QoL

This wave contains only vanilla-compatible and QoL UI mods. No gameplay-altering mechanics are included.

UI & Quality of Life

Better Camera Mod

Version: v8.1 (Jul 2026)

Impact: Unlocks closer and further camera zoom heights in battles and on the campaign map, and lets you zoom without units popping out of view. Makes battles feel more cinematic and easier to watch. Purely camera QoL with no gameplay change.

Cpecific's Skill Queue: Jazzed Edition

Version: v8.1 (Jul 2026)

Dependencies: Memreader (win64)

Impact: Lets you build skill templates and auto-assign skill points to every new lord and hero as they level up — never hand-click a skill tree again. The community-maintained fork of the original Skill Queue, with saved-game and faction templates plus auto-granted-skill detection. Saves hours; gives no gameplay advantage.

Memreader (win64)

Version: v8.1

Impact: Required memory-access library for Skill Queue and similar QoL mods. Windows x64 only. Install before any mod that lists it as a dependency.

The Tooltip Project

Version: v8.1 (Jul 2026)

Impact: Rewrites unit and spell tooltips so they tell you what actually matters: collision attacks, melee regeneration, hidden stat mechanics, and whether a buff spell is single- or multi-target. Information-only UI clarity — the author states it will never touch gameplay.

Building Progression Icons III

Version: v8.1 (Dec 2025)

Impact: Replaces flat building icons with artwork that visibly changes as each building is upgraded, so settlement state is readable at a glance. Purely cosmetic.

Speedy Turn Camera

Version: v8.1 (May 2026)

Impact: Speeds up the camera pan between factions during the AI's turn, so long end turns feel much faster. Changes a single database value — it does not speed up AI processing, just the camera travel.

High Resolution UI Improvements, YES IT STILL WORKS

Version: v8.1 (Dec 2025)

Impact: Resizes the recruitment, diplomacy, lord/hero, objectives, army-view and custom-battle panels so they display correctly on 1440p+ monitors. Note: some users report the recruitment panel can lag or freeze on 3440x1440 ultrawide.

Skip Intro

Version: v8.1 (May 2026)

Impact: Skips the startup intro videos and the epileptic-seizure warning at launch. Small convenience; be aware that running any mod adds a "Modified Game" confirmation popup at launch.

Notes

Wave 0 is designed for new players. The mods here only improve the interface and quality of life — nothing changes core game balance or mechanics. Skill Queue requires Memreader; install both or the preset features will misbehave.

Wave 0 — Graphics Mods

Base visual enhancements that improve the look of vanilla Warhammer III across all races and factions.

Colourless' Radious HD Texture Pack

Version: v8.1 (July 24 2026)

Impact: Comprehensive HD texture replacement covering Empire, Dwarfs, Greenskins, Vampire Counts, Bretonnia, Wood Elves, Beastmen, and more. Sharper unit details, improved armour textures, and cleaner faction heraldry with minimal performance impact.

Warhammer 1 Texture Update Project

Version: v8.1

Impact: Upgrades low-resolution textures for races carried over from Warhammer 1 — brings Empire State Troops, Dwarf Warriors, Greenskin Boyz, and other legacy units up to Warhammer 3 visual standards.

Campaign Map Lighting

Version: v8.1

Impact: Improves campaign map lighting with warmer tones, deeper shadows, and better contrast between day and night cycles. Enhances snow, desert, and jungle region visuals.

Notes

All Wave 0 graphics mods are purely cosmetic. No mechanical or gameplay impact. Mods are compatible with all DLC and other visual overhauls in later waves.

Wave 0 — Content Mods

Light, lore-friendly content additions that expand the vanilla experience without disrupting balance. Nothing here adds new factions, units, or mechanics — these enrich the world you already know.

Legendary Lore

Version: v8.1 (May 2026)

Impact: Adds lore-flavoured narrative text to the campaign: travel to significant locations to learn your lord's story, conquer racial capitals to trigger loreful events, and discover regional lore by building settlements and recruiting units. No gameplay impact whatsoever — pure flavour for the Old World.

Recruit Defeated Legendary Lords

Version: v8.1 (Jul 2026)

Impact: When a faction is destroyed, its legendary/immortal lords and heroes become available to surviving factions of the same race (player preferred, AI also benefits). Adds a refugee dilemma so you choose who joins you. A small campaign convenience that prevents lost lords from vanishing forever.

Notes

Content mods in Wave 0 add flavour and light conveniences. They do not introduce new mechanics or faction-specific systems — heavy content arrives with Wave 1.

Wave 0 — Mechanics Mods

> Wave 0 does not include any mechanics mods. Mechanics overhaul mods are introduced starting with Wave 2, when you have mastered the base game.

Leave this section empty for Wave 0.

Wave 1: The Gathering Storm

Roleplaying Background

The northern provinces burn. Word spreads that the Greenskins under Warboss Gorgutz have unified for the first time in a generation. The Dwarfs pull back from their crumbling Holds, and the Tomb Kings send scouts into the abandoned lands. You are no longer a minor Governor — you are a Warlord commanding the attention of factions that once ignored you.

Your campaign is no longer about survival. It is about ambition. You must secure alliances, claim ancient relics, and prepare the forces needed to face what comes next.

Wave 1 Strategy Guide — The Gathering Storm

Campaign Philosophy

Wave 1 introduces content mods that add new factions, units, and campaign features. Your strategy must adapt:

- **Expand faster.** New factions bring new threats and opportunities. Secure contested regions before they do.
- **Diplomacy is temporary.** Alliances shift as content packs introduce political mechanics. Never rely on a single ally.
- **Scout relentlessly.** Content mods add new regions and map features. Intelligence is your greatest asset.

Faction Strategy Adjustments

Empire

With the addition of Empire sub-factions and new units:

- Specialize: pick a sub-faction playstyle early (religious, martial, or diplomatic).
- Invest in the new hero options available through content mod additions.

Other Factions

Opposing factions also gain content bonuses. Expect:

- Stronger AI armies
- New AI expansion patterns
- Modified diplomatic behavior

Battle Tactics

New content changes the battlefield. Key adjustments:

1. **Identify new unit types.** Content mods may add units that counter existing strategies.
2. **Adapt your army composition.** Do not rely on a single build from Wave 0.
3. **Use terrain more aggressively.** New battlemap features may offer tactical advantages.

Economy in Wave 1

Content mods may alter trade mechanics. Watch for:

- New trade goods from content-added provinces
- Modified resource costs for new units
- Changing upkeep values for expanded armies

Readiness for Wave 2

You should be comfortable with:

- Managing expanded armies (3-4 armies simultaneously)
- Navigating new UI elements from content mods
- Adapting tactics mid-battle
- Balancing economy with rapid expansion

Wave 1 — Modlist: Content

This wave expands the game with new factions, units, legendary characters, and campaign features. Everything in this wave builds on the MIXER startpos framework, so install MIXER first and keep every dependency chain active.

Required Framework

MIXER - Mixu's Unlocker

Version: v8.1 (Jul 2026)

Dependencies: Mod Configuration Tool (recommended)

Impact: The startpos unlocker that makes all factions playable (458 lords in Immortal Empires) and the modder framework nearly every content mod in this wave is built on. Activate this first — content mods will not function without it. Note: single-player only for now; Aug 2026 CA update broke multiplayer.

New Factions

Cataph's Southern Realms (TEB) [It Works in 8.1]

Version: v8.1 (Dec 2025)

Dependencies: MIXER - Mixu's Unlocker; free WH3 Jade/Grey/Amethyst Wizard packs

Impact: Turns Tilea, Estalia, the Border Princes and New World Colonies into a full culture: 8 playable factions, 10 legendary lords and 1 legendary hero, 30+ new units, custom tech and building trees, landmarks, and the Plutocracy and mercenary-camp mechanics. Not for the Realms of Chaos campaign.

OvN Lost Factions: Fimir

Version: v8.1 (Jul 2026)

Dependencies: MIXER - Mixu's Unlocker (required); not multiplayer-campaign compatible

Impact: Adds the Fimir as a complete new playable race: full custom roster, 2 playable factions (Tendrils of Doom, Rancor Hold), and custom mechanics including Godhood Ascension, Gifts of Chaos and the Death Quest.

Cataph's Kraka Drak: the Norse Dwarfs [It Works in 8.1]

Version: v8.1 (Dec 2025)

Dependencies: MIXER - Mixu's Unlocker (required)

Impact: Adds the Norse Dwarf sub-faction under Great King Thorgard Cromson: 11+ new units, 8 Regiments of Renown, a custom legendary lord type, and new tech tree, buildings and landmarks. For Immortal Empires / The Old World campaigns.

New Legendary Lords & Characters

Mixu's Legendary Lords

Version: v8.1 (Jul 2026)

Dependencies: Mixu's Legendary Lords: Asset Pack

Impact: The community-standard legendary lord pack: 50+ lore-accurate legendary lords and heroes across existing races, with custom skill trees and new start positions. MIXER is recommended to play the added lords.

Mixu's Legendary Lords: Asset Pack

Version: v8.1 (Jul 2026)

Impact: Required asset pack for Mixu's Legendary Lords — without it you will get missing character models and other broken assets. Install alongside the main mod.

[DEER24] Walk with the Dragon (Cathay Units Patch)

Version: v8.1 (Aug 2026)

Dependencies: Mixu's faction unlocker (required); optional English-patch submod

Impact: Cathay mega-pack: 14 legendary lords, 11 legendary heroes, 2 new hero types, 60+ new units and a 50-turn event chain. Chinese-language text by default (English patch submod exists). Immortal Empires only.

Notes

Wave 1 mods add significant content. MIXER is the mandatory hub — activate it before anything else. Mixu's Legendary Lords requires its Asset Pack. All mods here are single-player focused; check conflicts.conf for startpos interactions (e.g. Immortal Empires Expanded + Landmarks of Eternity pairing is covered in the content section).

Wave 1 — Graphics Mods

Enhanced visual overhauls that add depth and detail to battles and campaign maps.

ULTIMATE LIGHTING 2.0

Version: v2.0

Impact: Complete battle lighting overhaul — deeper shadows, realistic sun positioning, improved ambient occlusion, and enhanced fog effects. Dramatically improves battle atmosphere across all climate types.

Details Matter: True Grit Textures (The Empire)

Version: v8.1

Impact: High-detail texture upgrade for Empire units — adds realistic wear, dirt, and battle damage to armour and clothing. Gritty visual style fitting the End Times setting.

Lore Soldiers: Units Reskin as Miniatures and Books

Version: v8.1

Impact: Reskins units across multiple factions to match tabletop Warhammer miniature aesthetics and official art. Covers High Elves, Dark Elves, Lizardmen, and more.

Notes

Wave 1 graphics mods are designed to complement each other. ULTIMATE LIGHTING 2.0 works alongside texture mods without conflict.

Wave 1 — Content Mods: World & Characters

These content mods expand the campaign world itself — new lands to conquer, landmarks to build, and a deeper cast of characters. They pair with the faction mods in the Wave 1 modlist and all share the MIXER framework.

Immortal Empires Expanded

Version: v8.1 (Jul 2026)

Dependencies: MIXER - Mixu's Unlocker (required); incompatible with other startpos mods

Impact: Expands the Immortal Empires map with 184 new settlements, 74 provinces and 55 factions across Ind, Khuresh, the Lost Isles of Elithis, Eastern Steppes, Khosun and Nippon. Fills the empty map edges with vanilla-plus world content.

Landmarks of Eternity

Version: v8.1 (May 2026)

Dependencies: MIXER recommended; some landmarks require it

Impact: Adds 420+ new landmarks to Immortal Empires (290 for the Old World) with unique dilemmas, items, icons and roughly 50 Regiments of Renown. Has an official "Landmarks of Eternity IEE Edition" submod for pairing with Immortal Empires Expanded.

Legendary Characters - 8.1

Version: v8.1 (Aug 2026)

Dependencies: Mod Configuration Tool (required)

Impact: Adds 103 lore characters (legendary lords and heroes) with custom skill trees, quest items and some units/factions. Every character can be toggled on/off via the Mod Configuration Tool. Complements Mixu's Legendary Lords without overlapping it.

Notes

World content in Wave 1 significantly expands the campaign map and character roster. Ensure all dependencies are met before activation — MIXER first, then the content chain. For Landmarks of

Eternity + Immortal Empires Expanded, use the official IEE Edition submod (id 3079915520) instead of the base version.

Wave 1 — Mechanics Mods

> Wave 1 does not yet include deep mechanics overhauls. Mechanics mods are introduced starting with Wave 2, when you have a solid grasp of the core game.

Leave this section empty for Wave 1.

Wave 2: The End Times

Roleplaying Background

The Old World is dying. From the north comes a tide of corruption that devours everything it touches. The Empire's borders have collapsed. The Dwarf Holds are silent. Even the Elves have abandoned their shores.

You are what remains. A warlord of a fractured realm, commanding the last loyal soldiers in a world that has forgotten hope. Your fortress is the only bastion of order for a hundred leagues. Every day brings new threats — not just from the armies of Chaos, but from within. Resources dwindle. The gods have abandoned you.

But you did not come this far to die.

Wave 2 Guide — The End Times

Campaign Philosophy

Wave 2 introduces SFO: Grimhammer III and the mechanics mods built around it. These fundamentally change how you play. Your approach must evolve:

- **SFO pacing is the new normal.** Battles last longer, XP comes slower, and every unit has a defined role. SFO rebalances the entire game around its lore-flavoured vision — expect to relearn matchups you thought you knew.
- **Winning is a journey, not a checklist.** Victory Conditions Overhaul replaces short/long objectives with three thematic routes per faction (50–70 turns), each with unique rewards, quest battles and epilogues. Decide early which route you are pursuing — objectives are set on turn 1.
- **Armies must be built, not stacked.** Tabletop Caps limit Core/Special/Rare units per army for the player AND the AI. Doomstacks are gone; balanced, lore-like armies are the standard.
- **The AI plays smarter.** AI Recruitment & Army Compositions and the Siege AI Overhaul mean enemy armies are balanced and sieges are fought, not attrition-starved.

Core Mechanics Changes

SFO Combat Pacing

Battles are longer and more decisive. Units are slower to break and harder to replace.

- Protect your elite units — SFO makes losses painful and recovery expensive.
- Use terrain and flanking; brute-force trades favour the AI's numbers.

Victory Routes

Each faction has 3 thematic victory routes with unique rewards and quest battles.

- Choose your route by turn 1 and commit.
- Route rewards include custom units and epilogues — read the VCO objectives panel regularly.

Army Composition Limits

Tabletop-style caps force balanced armies for both sides.

- Plan rosters around Core (reliable), Special (situational) and Rare (decisive) slots.
- The AI operates under the same limits — exploit that symmetry.

Smarter AI

Enemy armies are themed around their legendary lord and sieges are aggressive.

- Expect AI armies with proper line infantry and capped monsters, not spam.
- Defend walls and gates — Siege AI rework makes towers and attrition meaningful.

Difficulty Curve

Wave 2 is designed to be punishing. Key challenges:

- Longer, more tactical battles under SFO
- AI that builds balanced, thematic armies
- Aggressive siege AI that attacks instead of waiting
- Victory routes that demand focused, long-term planning

Recommended Progression

- Turns 1-10: Consolidate your fortress. Pick your VCO route. Build toward the army cap structure you want.
- Turns 11-25: Expand carefully. Take only provinces you can defend against a smarter AI.
- Turns 26+: Begin offensive operations. Strike at weakened enemies and pursue your victory route.

Prerequisites Before Wave 3

You should be comfortable with:

- Managing SFO's longer battles and slower progression
- Building armies under Tabletop Caps
- Pursuing a VCO victory route
- Beating a balanced, aggressive AI

Important

Start a new campaign when activating Wave 2. SFO, VCO and Tabletop Caps require it — do not continue a Wave 1 save. Install the VCO-SFO compatibility patch and the SFO patches for any Wave 1 content mods you use (see the content and mechanics sections).

Wave 2 — Modlist: Mechanics

This wave introduces mechanics overhaul mods that change core gameplay systems. SFO: Grimhammer III is the foundation — every other mechanic in this wave is chosen to work with it. Start a new campaign when activating this wave.

Total Overhaul

SFO: Grimhammer III - Call of Bretonnia

Version: v8.1 (Jul 2026)

Dependencies: None required; fully standalone

Impact: The flagship lore-flavoured total overhaul: longer battles, faction-mechanic reworks, slower XP progression, optional unit caps, and everything rebalanced against the tabletop-inspired vision. Completely replaces vanilla balance — this is the single biggest change in the whole modlist.

Campaign Systems

Victory Conditions Overhaul

Version: v8.1 (Jul 2026)

Dependencies: VCO - SFO Compatibility Patch (required with SFO)

Impact: Replaces the short/long victory objectives with 3 thematic routes per faction (50–70 turns), each with unique rewards, quest battles and epilogues. Reshapes what "winning" means. Must be enabled at campaign start — objectives are set on turn 1.

Diplomacy Logic and Balance Katsuko

Version: v8.1 (Dec 2025)

Dependencies: None; stated compatible with all mods incl. SFO

Impact: Rebalances every core diplomatic factor: distance-aware war declarations, no instant trust between enemies, selling/gifting/purchasing settlements, and dynamic alliances. Diplomacy becomes logical and strategic instead of arbitrary.

Army & Battle

Tabletop Caps: Reborn

Version: v8.1 (Dec 2025)

Dependencies: Optional: Mod Configuration Tool for tuning

Impact: Tabletop-style Core/Special/Rare unit caps per army, applied to the player AND the AI. Kills doomstacks and forces balanced, lore-like army compositions. The primary army-cap system for this wave.

AI Recruitment & Army Compositions (8.1.1)

Version: v8.1.1 (Jul 2026)

Dependencies: None; incompatible with mods touching `cdir_military_generator`

Impact: Rewrites how the AI builds armies: more line infantry, capped ranged/artillery/monsters, and themed templates per legendary lord. AI fielding balanced armies instead of spam. Incompatible with Campaign AI Tweaks — do not run both.

Siege AI & Mechanics Overhaul

Version: v8.1 (May 2026)

Dependencies: None; compatible with most garrison mods

Impact: Reworks sieges: the AI attacks instead of attrition-starving, defensive buildings grant attrition/ammo/upkeep bonuses, no post-battle tower building, healthier gates and walls, and reworked settlement battle types.

Notes

Wave 2 mechanics mods are deep gameplay changes. Activate SFO first, then the rest. The VCO-SFO compatibility patch and SFO content patches for Wave 1 mods are listed in the mechanics and content

sections — read those before launching. Never pair SFO with Radious Total War (see conflicts.conf). Start a fresh campaign.

Wave 2 — Graphics Mods

Darkened visuals reinforcing the End Times apocalyptic setting.

Grim Lighting

Version: v8.1

Impact: Dark, oppressive battle lighting with reduced saturation and increased shadow density. Creates a grim atmosphere that suits the End Times — overcast skies, blood-drenched battlefields, and muted colour palette.

Chaos Visual Overhaul

Version: v8.1

Impact: Comprehensive visual overhaul for all Chaos factions — Warriors of Chaos, Daemons of Chaos, and Beastmen. Updated armour textures, corrupted banner designs, and Daemon model refinements.

Better Chaos Waste Lighting Effect

Version: v8.1

Dependencies: Chaos Visual Overhaul

Impact: Adds unique atmospheric lighting to Chaos Waste regions on the campaign map — eerie green glows, volatile skies, and corruption-based visual effects that intensify as Chaos influence spreads.

Notes

Wave 2 graphics mods reinforce the End Times atmosphere. Some are chained to other mods. Compatible with Waves 0-1 graphics.

Wave 2 — Content Mods: SFO Integration

The End Times setting is built on SFO: Grimhammer III, so all Wave 1 content must be rebalanced to SFO's pacing. These official-style compatibility patches (by Fesir) adapt the Wave 1 content mods to SFO combat, economy and unit-cap logic. Activate each patch for the content mods you carry over from Wave 1.

Mixu's Legendary Lords - SFO Compatibility Patch

Version: v8.1 (Jul 2026)

Dependencies: Mixu's Legendary Lords; Mixu's Legendary Lords: Asset Pack; SFO: Grimhammer III

Impact: Adapts Mixu's lords, heroes, mounts and units to SFO combat pacing, costs, recruitment and unit-cap logic. Load order (top to bottom): this patch, Mixu's Legendary Lords, Asset Pack, SFO.

Cataph's Southern Realms - SFO Compatibility Patch

Version: v8.1 (Aug 2026)

Dependencies: Cataph's Southern Realms (TEB); MIXER - Mixu's Unlocker; SFO: Grimhammer III

Impact: Rebalances the Southern Realms roster for SFO: unit stats, costs, upkeep, recruitment, Dogs of War unit capacities, buildings, skills and starting armies. Load order (top to bottom): this patch, TEB, SFO, MIXER.

Landmarks of Eternity — SFO Compatibility Patch

Version: v8.1 (Jul 2026)

Dependencies: Landmarks of Eternity; SFO: Grimhammer III; (optional) Landmarks of Eternity IEE add-on + Immortal Empires Expanded

Impact: Rebalances Landmarks of Eternity landmark units and Regiments of Renown for SFO's battle pacing and economy. Load order (top to bottom): this patch, LoE IEE add-on, Landmarks of Eternity, Immortal Empires Expanded, SFO, MIXER.

Notes

These patches are compatibility layers, not new content — they make Wave 1 content playable under SFO. Only install the patches for content mods you actually use. A new campaign is strongly recommended after enabling SFO. Do not stack multiple SFO patches for the same mod.

Wave 2 — Mechanics Mods (continued)

Additional mechanics that tie the Wave 2 systems together under SFO.

Victory Conditions Overhaul - SFO Compatibility Patch

Version: v8.1 (Jul 2026)

Dependencies: Victory Conditions Overhaul; SFO: Grimhammer III; (optional) Tabletop Caps: Reborn

Impact: Updates VCO rewards, reward units, effects and script support for current SFO balance while preserving the route-based campaign structure. Adds guarded Tabletop Caps support for VCO reward units. Load order (top to bottom): this patch, VCO, SFO, Tabletop Caps.

Notes

These mechanics mods are the most impactful in the modlist. They are designed to be used together but can be activated selectively if you want a lighter challenge.

The VCO-SFO patch is required whenever Victory Conditions Overhaul and SFO are active together. Do not use multiple VCO-SFO patches at once. A new campaign is recommended after any major update to VCO, SFO, Tabletop Caps or the game.

Wave 3: When Gods Bleed

Roleplaying Background

The gods bleed. Reality tears at the seams. You have seen empires rise and crumble, watched friends die and heroes fall. The world as you knew it is gone — replaced by a shattered landscape of warring realms and bleeding skies.

You are what remains of the old world's champions. Some call you the last hope. Others call you a fool for continuing. You do not answer to titles or gods anymore. You answer only to the blade in your hand and the blood beneath your feet.

This is the end. And only the strong — or the desperate — will survive.

Wave 3 — When Gods Bleed

Campaign Philosophy

Wave 3 is the culmination of the entire modlist. Every mod from Waves 0–3 is active simultaneously. There are no holds. No mercy. No undo.

- **Everything is active.** All mods from previous waves are in play.
- **No easy mode.** Difficulty is at its absolute maximum.
- **No resets.** Every loss is permanent. Every victory is hard-won.
- **Adapt or die.** The challenge is relentless and unrelenting.

Combined Mod Impact

With all waves active, the following compound effects exist:

1. **SFO + Victory Routes** — SFO's rebalanced pacing drives every campaign decision, while VCO routes give you a long-term goal worth fighting for.
2. **Content + Unit Caps** — New factions, lords and units from Wave 1 must be managed under Tabletop Caps. Fresh armies are fragile; elite units are precious.
3. **Smarter AI + Sieges** — AI Recruitment and the Siege AI Overhaul make every war a fight against balanced, thematic armies attacking your walls.
4. **Graphics + Atmosphere** — The full visual stack makes the campaign map a bleak, hostile place. No visual respite.

Difficulty Expectations

Wave 3 is designed to be the hardest possible campaign experience in TWWH3. Expect:

- Enemy factions with all content and mechanics bonuses active
- SFO-length battles where every mistake is punished
- No forgiveness for tactical errors
- Army compositions constrained by caps — for you and the AI alike
- A campaign that tests every system simultaneously

Recommended Approach

1. Start in the most defensible position possible.
2. Commit to a VCO victory route early — a faction without direction is a faction that dies.
3. Build armies around the cap structure; do not waste Rare slots on redundancy.
4. Economy first, military second. You cannot afford a standing army you cannot maintain.
5. Accept that losses are inevitable. Focus on managing the rate of loss, not preventing it.

6. Keep save files religiously before major battles.

Final Note

Wave 3 is for veterans only. If you have not completed at least Wave 2 successfully, return to it. The gods do not bleed easily — and neither should you die easily.

Wave 3 — Modlist: Full Chaos

This wave activates ALL mods from Waves 0–3 simultaneously. This is the complete modlist. Activate in wave order: QoL (below), graphics (below + Wave 3 graphics section), then content and mechanics from the Wave 1/2 sections.

UI / QoL (Wave 0 Carryover)

Better Camera Mod

Version: v8.1 (Jul 2026)

Impact: Unlocked camera heights in battles and campaign; zoom without unit culling.

Cpecific's Skill Queue: Jazzed Edition

Version: v8.1 (Jul 2026)

Dependencies: Memreader (win64)

Impact: Skill presets that auto-assign points to new lords/heroes.

Memreader (win64)

Version: v8.1

Impact: Memory library required by Skill Queue.

The Tooltip Project

Version: v8.1 (Jul 2026)

Impact: Rewrites unit/spell tooltips with hidden stats and mechanics.

Building Progression Icons III

Version: v8.1 (Dec 2025)

Impact: Building icons change as buildings upgrade.

Speedy Turn Camera

Version: v8.1 (May 2026)

Impact: Faster end-turn camera pan between factions.

High Resolution UI Improvements

Version: v8.1 (Dec 2025)

Impact: Resizes UI panels for 1440p+ monitors.

Skip Intro

Version: v8.1 (May 2026)

Impact: Skips startup intros and the epilepsy warning.

Graphics (Waves 0–2 Carryover)

Colourless' Radious HD Texture Pack

Version: v8.1 (Jul 2026)

Impact: Comprehensive HD texture replacement across all base races.

Warhammer 1 Texture Update Project

Version: v8.1

Impact: Upgrades WH1 legacy race textures to WH3 standards.

Campaign Map Lighting

Version: v8.1

Impact: Warmer campaign map lighting with deeper shadows and day/night contrast.

ULTIMATE LIGHTING 2.0

Version: v2.0

Impact: Complete battle lighting overhaul with deeper shadows and better ambient occlusion.

Details Matter: True Grit Textures (The Empire)

Version: v8.1

Impact: High-detail Empire texture upgrade with realistic wear and battle damage.

Lore Soldiers: Units Reskin as Miniatures and Books

Version: v8.1

Impact: Tabletop-miniature-accurate reskins across multiple factions.

Grim Lighting

Version: v8.1

Impact: Dark, oppressive battle lighting with reduced saturation — the End Times look.

Chaos Visual Overhaul

Version: v8.1

Impact: Comprehensive visual overhaul for all Chaos factions.

Better Chaos Waste Lighting Effect

Version: v8.1

Dependencies: Chaos Visual Overhaul

Impact: Unique atmospheric lighting for Chaos Waste regions.

Notes

Wave 3 is the final wave. All mods from Waves 0–3 are active. Respect all dependency chains: Skill Queue needs Memreader, VCO needs its SFO patch, Wave 1 content needs MIXER and its SFO patches. Check conflicts.conf before activation — some mods may conflict even if not within the same wave.

Wave 3 — Graphics Mods

Faction-specific texture upgrades and visual polish for individual races.

Dark Elves Dark Textures

Version: v8.1

Impact: Darkens and sharpens Dark Elf unit textures — deeper blacks on armour, richer crimson accents on cloaks, and more defined scale patterns on monsters. Fits the darker tone of the End Times.

Dwarf Lore Reskin (Visual Overhaul Compilation)

Version: v8.1

Impact: Updated Dwarf unit models with lore-accurate helmet designs, shield heraldry, and armour variants. Covers Dwarf Warriors, Longbeards, Ironbreakers, Hammerers, and Slayers.

Tomb Kings Reskin (Lore Visual Overhaul)

Version: v8.1

Impact: Visual overhaul for Tomb Kings — reshaded constructs, refined hieroglyphic details on war machines, and improved Lord/Hero model fidelity.

3D Porthole Lighting Fix

Version: v8.1

Impact: Fixes the flat lighting on 3D character portraits in the campaign UI. Restores proper shading and specular highlights to lord and hero porthole portraits.

Notes

Wave 3 adds race-specific visual detail on top of the base texture upgrades from Waves 0-2. All mods are purely cosmetic with no gameplay impact.

Wave 3 — Content Mods

All content mods are active. No additional content mods are needed — Wave 3 carries over all content from Waves 0–2.

See Wave 0 Content, Wave 1 Content, and Wave 2 Content sections for the full list of active content mods.

Wave 3 — Mechanics Mods

All mechanics mods are active. No additional mechanics mods are needed — Wave 3 carries over all mechanics from Wave 2.

See Wave 2 Mechanics sections for the full list of active mechanics mods.

Final Note

In Wave 3, every mechanics mod is running simultaneously. The compound effect creates the most challenging experience possible. No single mod is balanced in isolation — balance only exists with the full stack active.